Ridge Regression:

Regulating overfitting when using many features

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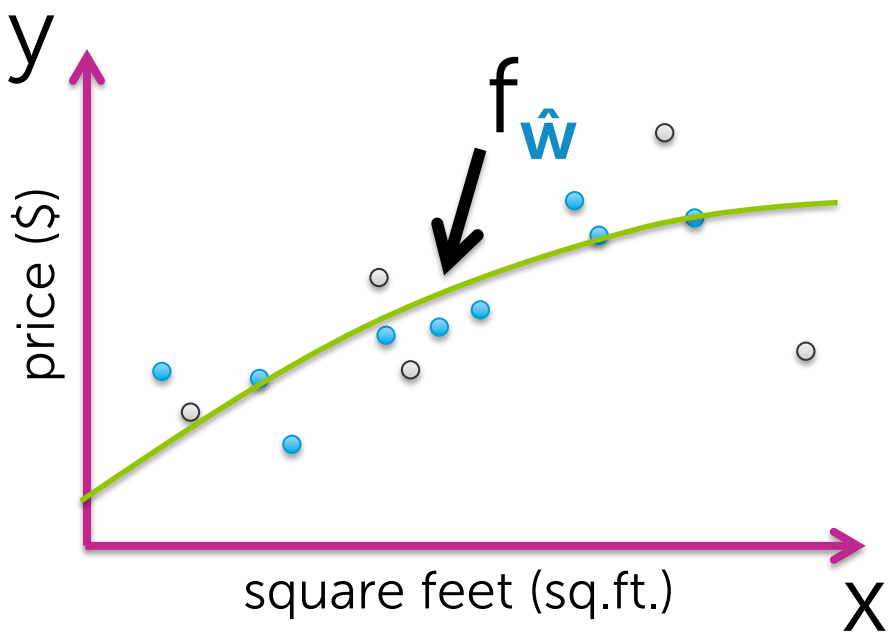
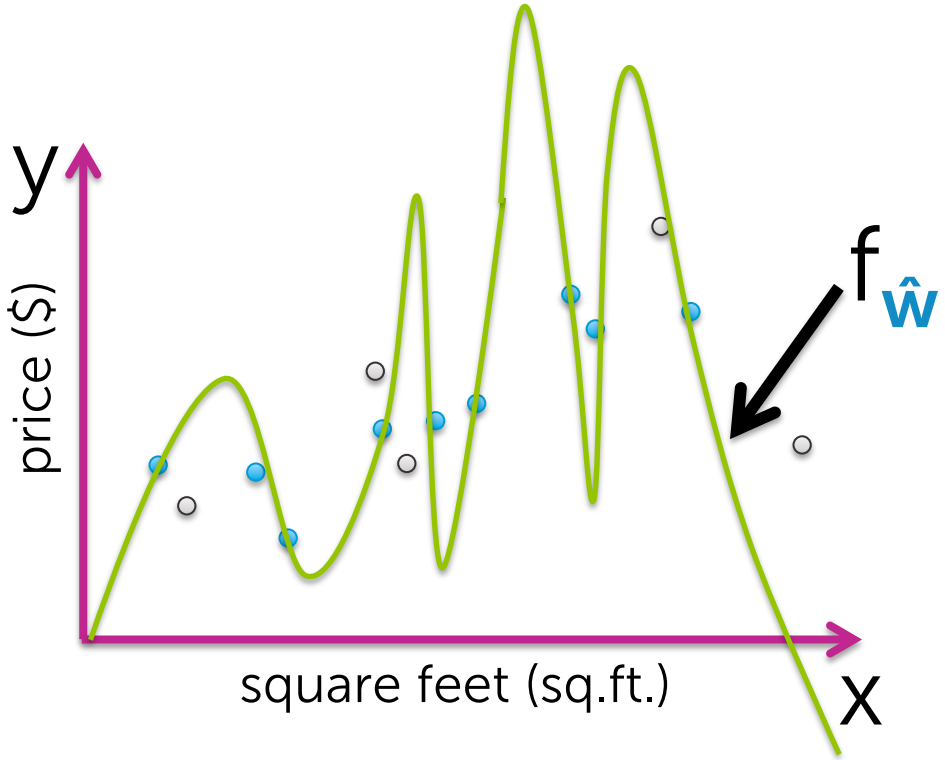
Overfitting of polynomial regression

Flexibility of high-order polynomials

$$y_i = w_0 + w_1 x_i + w_2 x_i^2 + \ldots + w_p x_i^p + \varepsilon_i$$





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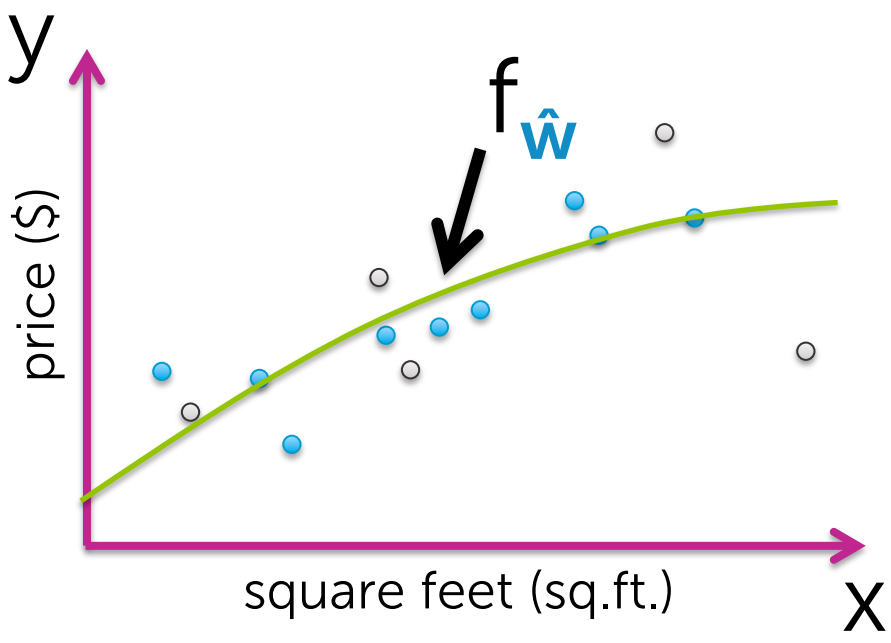
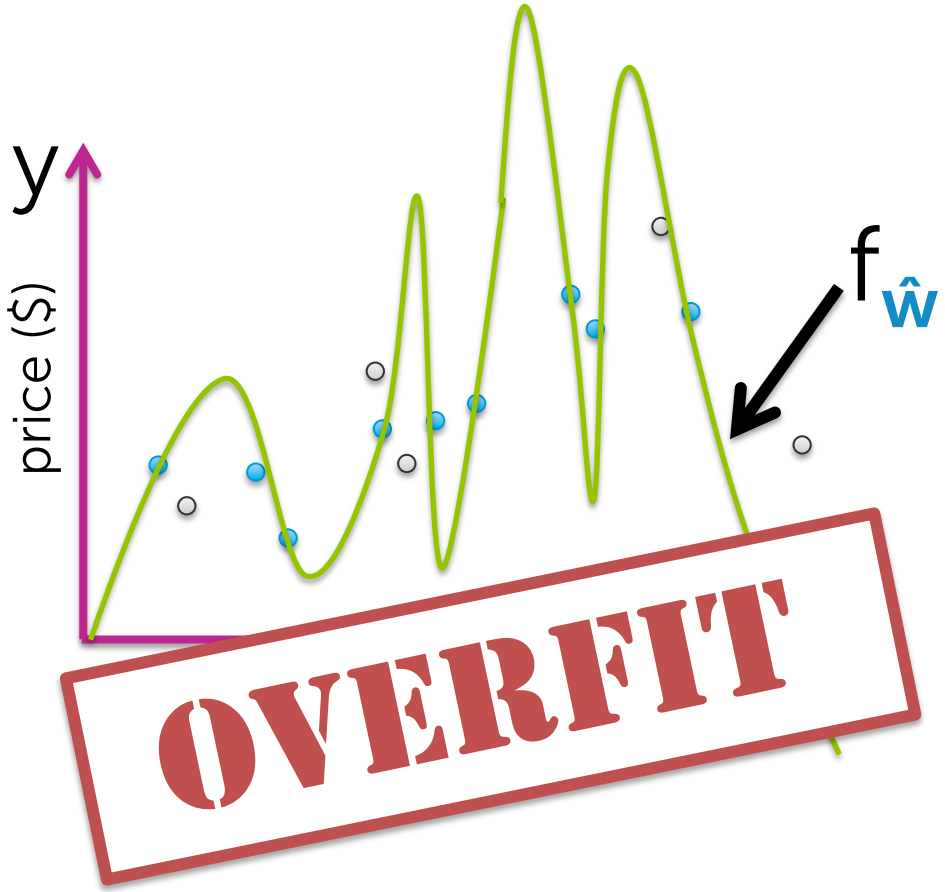
Flexibility of high-order polynomials

$$y_i = w_0 + w_1 x_i + w_2 x_i^2 + \ldots + w_p x_i^p + \varepsilon_i$$





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Symptom of overfitting

Often, overfitting associated with very large estimated parameters $\hat{\mathbf{w}}$

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Overfitting of linear regression models more generically

Overfitting with many features

Not unique to polynomial regression,
but also if **lots of inputs** (d large)

- Square feet
- # bathrooms
- # bedrooms
- Lot size
- Year built
- …

Or, generically,
lots of features (D large)

$$y_i = \sum_{j=0}^{D} w_j \, h_j(\mathbf{x}_i) + \varepsilon_i$$

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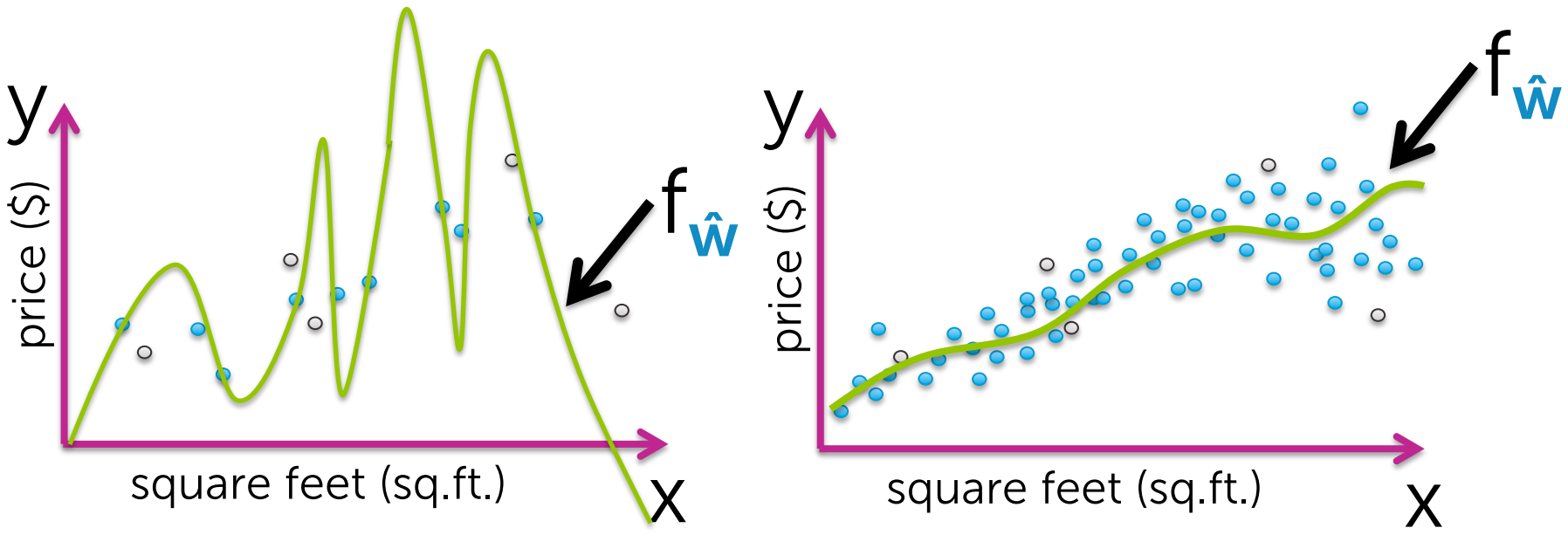
How does # of observations influence overfitting?

Few observations (N small)
→ rapidly overfit as model complexity increases

Many observations (N very large)
→ harder to overfit



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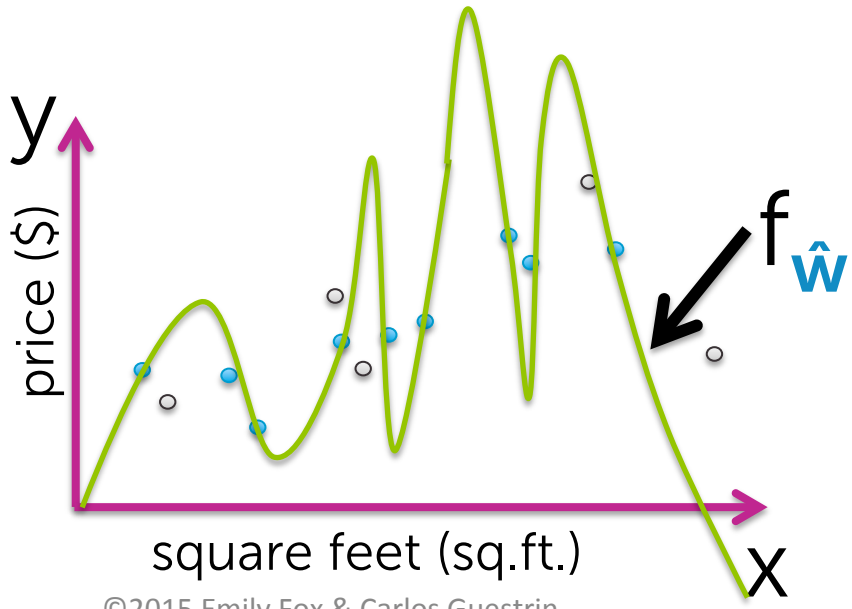
How does # of inputs influence overfitting?

1 input (e.g., sq.ft.):

Data must include representative examples of all possible (sq.ft., $) pairs to avoid overfitting



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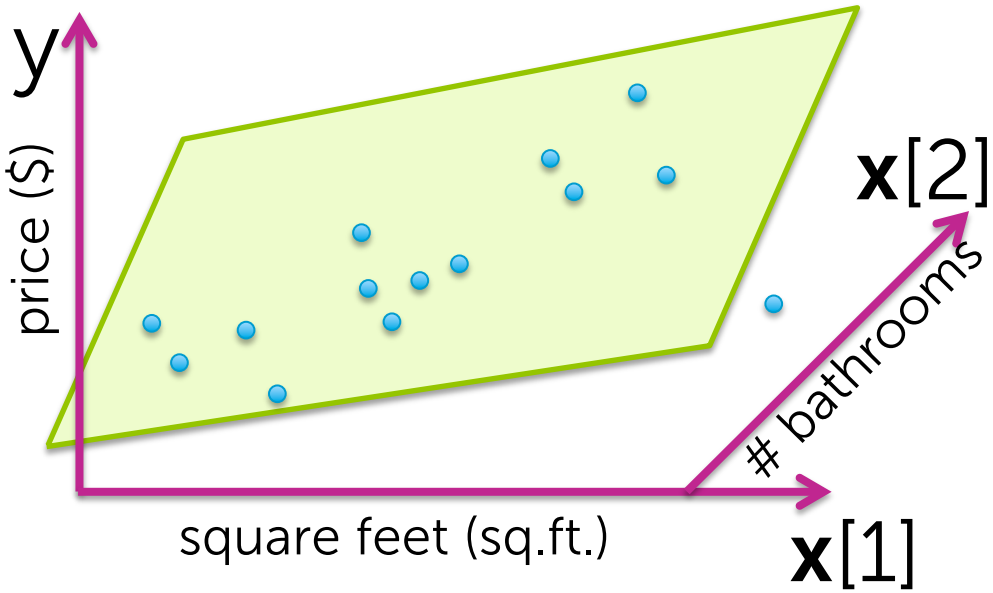
How does # of inputs influence overfitting?

d inputs (e.g., sq.ft., #bath, #bed, lot size, year,…):

Data must include examples of all possible (sq.ft., #bath, #bed, lot size, year,…., $) combos to avoid overfitting



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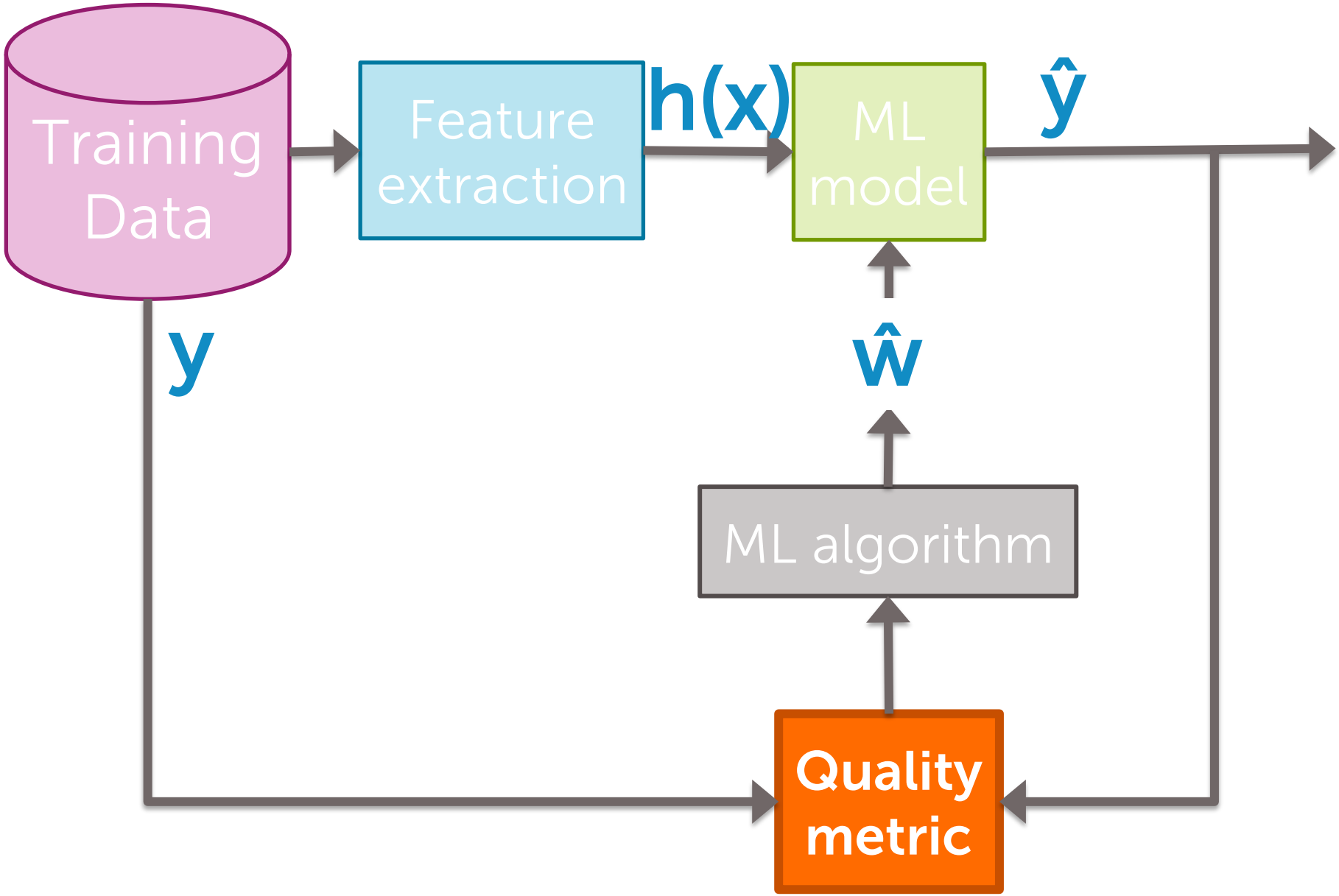
Adding term to cost-of-fit to prefer small coefficients



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Desired total cost format

Want to balance:

i. How well function fits data

ii. Magnitude of coefficients

want to balance

measure quality of fit

Total cost =

measure of fit + measure of magnitude of coefficients

small # = good fit to training data

small # = not overfit

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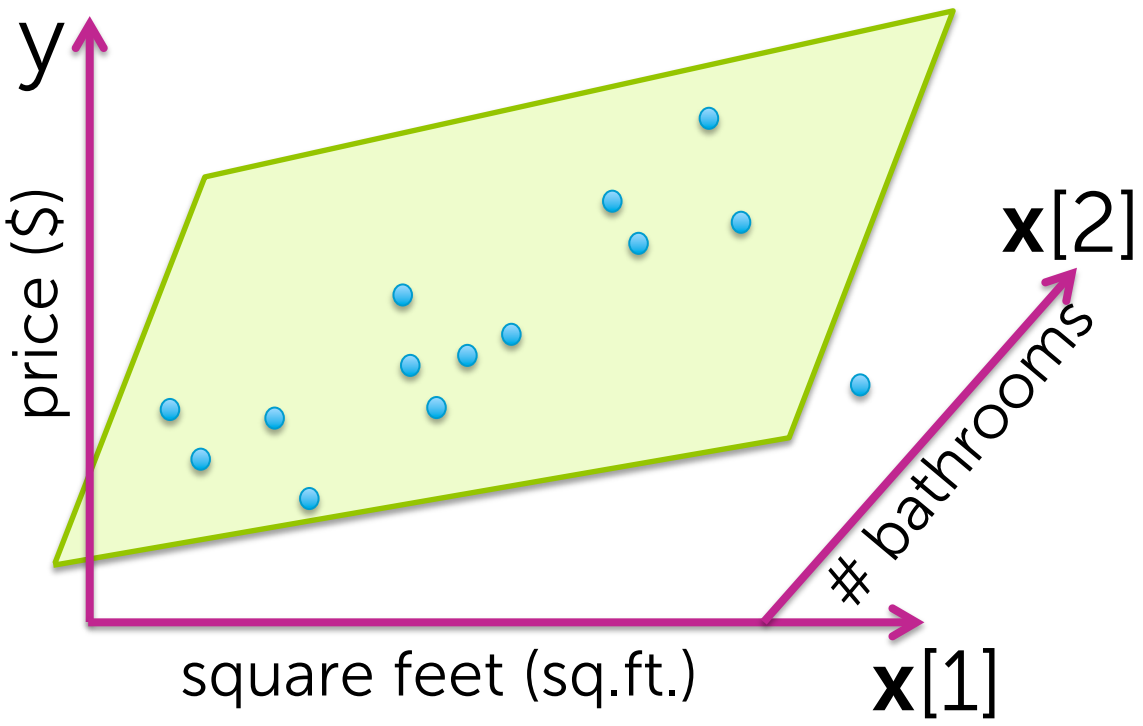
Measure of fit to training data



$$\text{RSS}(\mathbf{w}) = \sum_{i=1}^{N} (y_i - h(\mathbf{x}_i)^T \mathbf{w})^2$$

$$= \sum_{i=1}^{N} (y_i - \hat{y}_i(\mathbf{w}))^2$$

pred. value using w

small RSS → model fitting training data well

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Measure of magnitude of regression coefficient

What summary # is indicative of size of regression coefficients?

- Sum? $w_0 = 1{,}527{,}301 \quad w_1 = -1{,}605{,}253$

  $w_0 + w_1 = \text{small } \#$

- Sum of absolute value?

  $|w_0| + |w_1| + \ldots + |w_D| = \sum_{j=0}^{D} |w_j| \triangleq \|\mathbf{w}\|_1$ $\quad L_1$ norm ... discuss more in next module

- Sum of squares (L_2 norm)

  $w_0^2 + w_1^2 + \ldots + w_D^2 = \sum_{j=0}^{D} w_j^2 \triangleq \|\mathbf{w}\|_2^2$ $\quad L_2$ norm ... focus of this module

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Consider specific total cost

Total cost =

measure of fit + measure of magnitude of coefficients

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Consider specific total cost

Total cost =

$$\underbrace{\text{measure of fit}}_{\text{RSS}(\mathbf{w})} + \underbrace{\text{measure of magnitude of coefficients}}_{||\mathbf{w}||_2^2}$$

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Consider resulting objective

What if $\hat{\mathbf{w}}$ selected to minimize

$$\text{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|_2^2$$

tuning parameter = balance of fit and magnitude

If $\lambda=0$:

reduces to minimizing RSS(w), as before (old solution) $\rightarrow \hat{w}^{LS}$ ← least squares

If $\lambda=\infty$:

For solutions where $\hat{w} \neq 0$, then total cost is ∞

If $\hat{w}=0$, then total cost = RSS(0) $\longrightarrow$ solution is $\hat{w}=0$

If λ in between: Then $0 \leq \|\hat{w}\|_2^2 \leq \|\hat{w}^{LS}\|_2^2$

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Consider resulting objective

What if $\hat{\mathbf{w}}$ selected to minimize

$$\text{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|_2^2$$

tuning parameter = balance of fit and magnitude

Ridge regression
(a.k.a L_2 regularization)

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Bias-variance tradeoff

Large λ:

high bias, low variance

(e.g., $\hat{\mathbf{w}} = 0$ for $\lambda = \infty$)

In essence, λ controls model complexity

Small λ:

low bias, high variance

(e.g., standard least squares (RSS) fit of high-order polynomial for $\lambda = 0$)

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Revisit polynomial fit demo

What happens if we refit our high-order polynomial, but now using **ridge regression**?

Will consider a few settings of λ …

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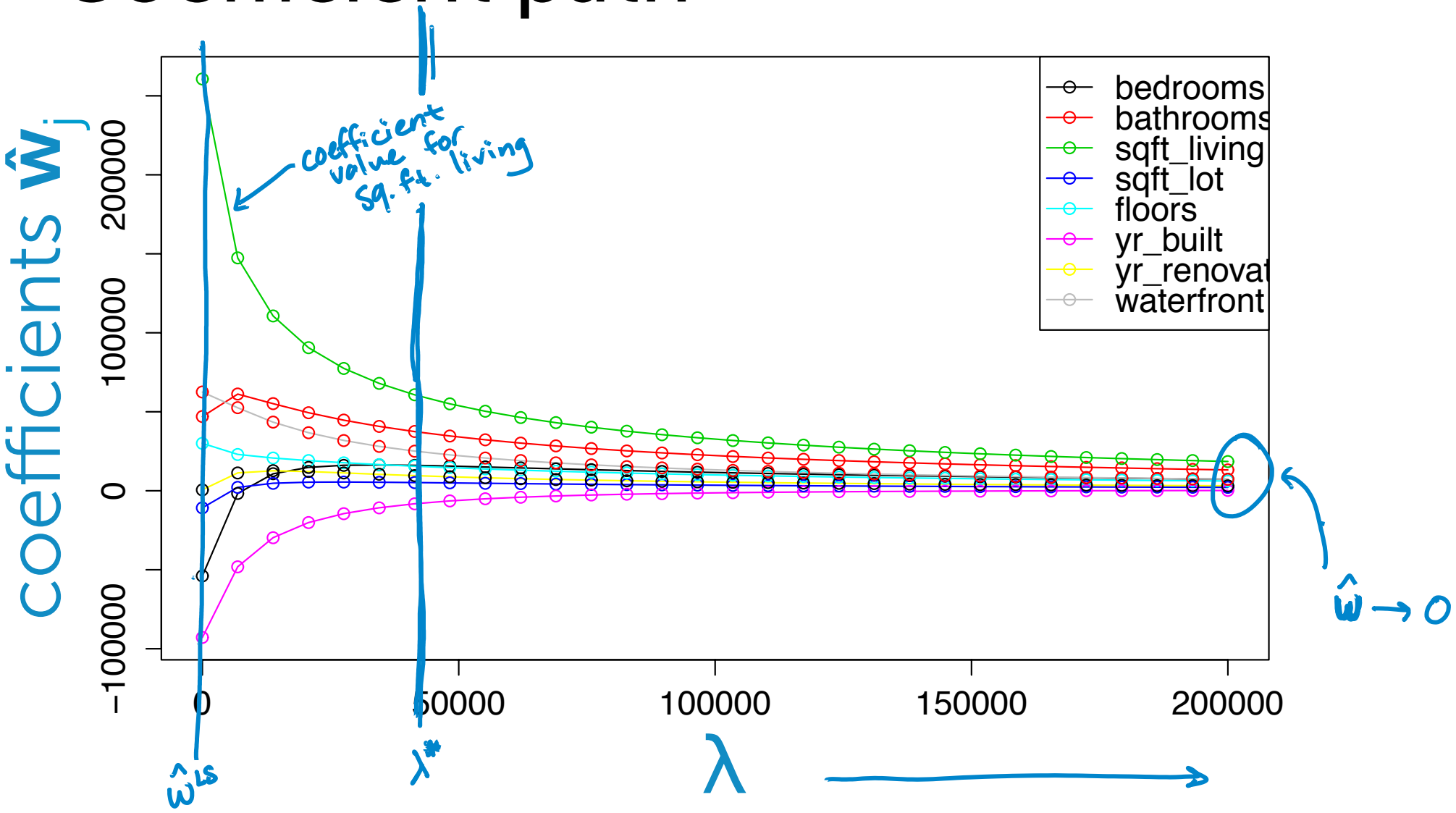
Coefficient path



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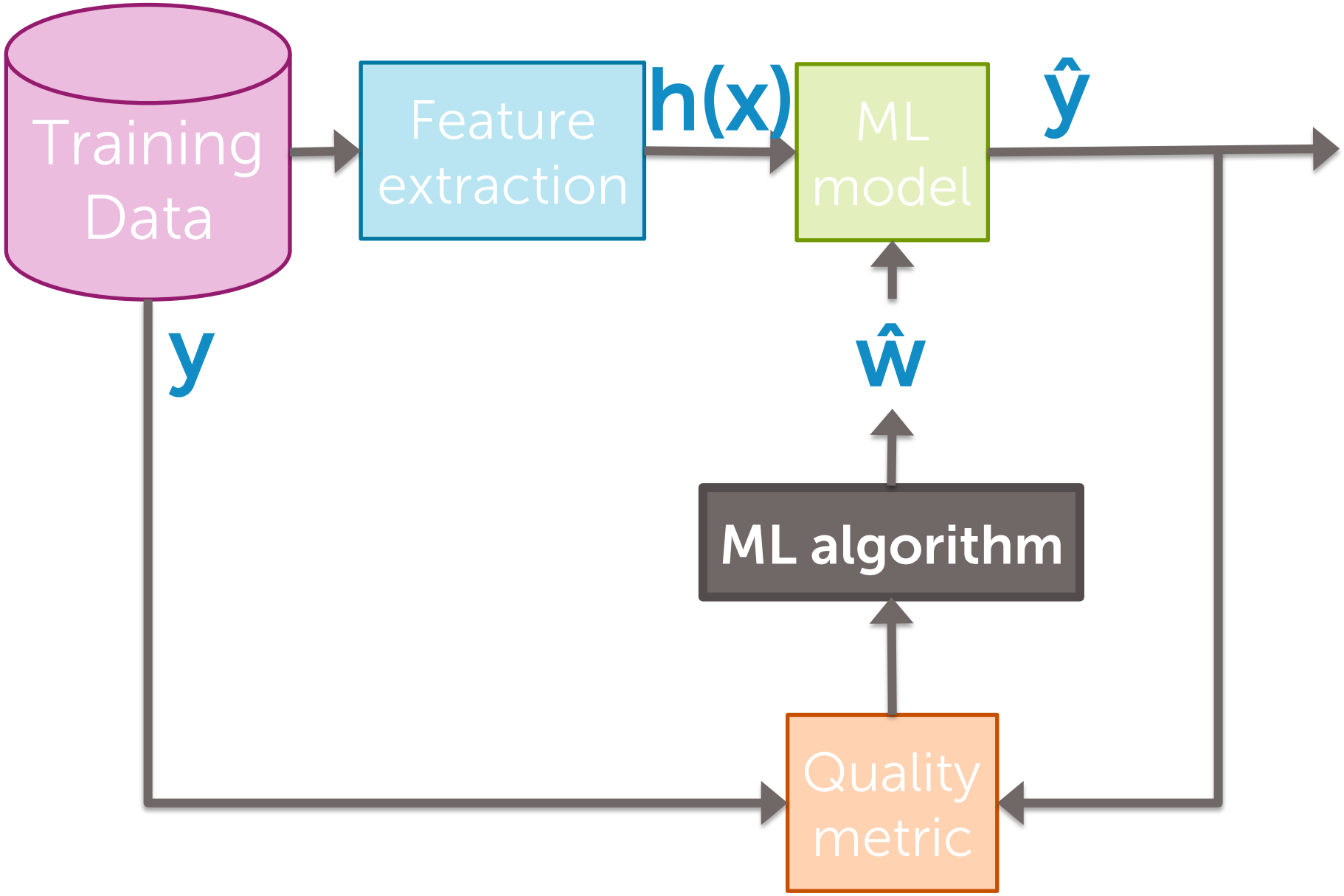
Fitting the ridge regression model (for given λ value)



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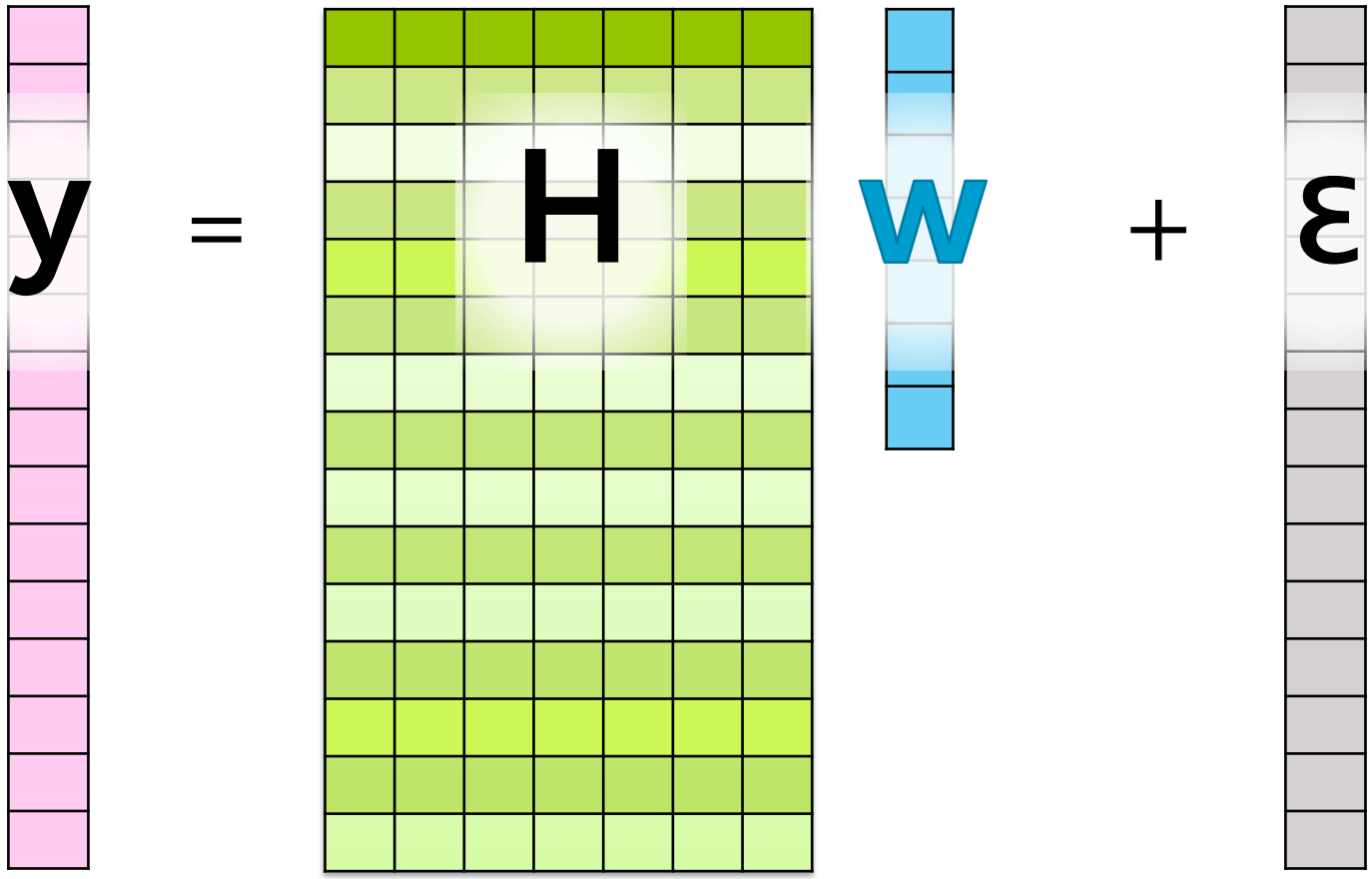
Step 1:
Rewrite total cost in matrix notation

Recall matrix form of RSS

Model for all N observations together



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Recall matrix form of RSS

$$\text{RSS}(\mathbf{w}) = \sum_{i=1}^{N} (y_i - h(\mathbf{x}_i)^T \mathbf{w})^2$$

$$= (\mathbf{y} - \mathbf{H}\mathbf{w})^T (\mathbf{y} - \mathbf{H}\mathbf{w})$$

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Rewrite magnitude of coefficients in vector notation

$$||\mathbf{w}||_2^2 = w_0^2 + w_1^2 + w_2^2 + \ldots + w_D^2$$

$$= \begin{array}{c} \begin{array}{ccccccc} \square & \square & \square & \square & \square & \square & \square \end{array} \\ \begin{array}{ccccccc} w_0 & w_1 & w_2 & \cdots & & & w_D \end{array} \end{array} \begin{array}{c} \square \\ \square \\ \square \\ \square \\ \square \\ \square \\ \square \end{array} \begin{array}{l} w_0 \\ w_1 \\ w_2 \\ \vdots \\ \\ \\ w_D \end{array}$$

$$= \mathbf{w}^T \mathbf{w}$$

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Putting it all together

In matrix form, ridge regression cost is:

$$\text{RSS}(\mathbf{w}) + \lambda||\mathbf{w}||_2^2$$

$$= (\mathbf{y}-\mathbf{H}\mathbf{w})^T(\mathbf{y}-\mathbf{H}\mathbf{w}) + \lambda\mathbf{w}^T\mathbf{w}$$

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Step 2:
Compute the gradient

Gradient of ridge regression cost

$$\nabla[\mathrm{RSS}(\mathbf{w}) + \lambda||\mathbf{w}||_2^2] = \nabla[(\mathbf{y}-\mathbf{H}\mathbf{w})^T(\mathbf{y}-\mathbf{H}\mathbf{w}) + \lambda\mathbf{w}^T\mathbf{w}]$$

$$= \underbrace{\nabla[(\mathbf{y}-\mathbf{H}\mathbf{w})^T(\mathbf{y}-\mathbf{H}\mathbf{w})]}_{-2\mathbf{H}^T(\mathbf{y}-\mathbf{H}\mathbf{w})} + \lambda \underbrace{\nabla[\mathbf{w}^T\mathbf{w}]}_{2\mathbf{w}}$$

Why? By analogy to 1d case…

$\mathbf{w}^T\mathbf{w}$ analogous to w^2 and derivative of $w^2=2w$

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Step 3, Approach 1:
Set the gradient = 0

Aside:
Refresher on identity matrics

$$I_1 = [1],\ I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},\ I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},\ \cdots,\ I_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

Fun facts:

$$\overset{n\times n}{\mathbf{I}}\underset{\text{vector}}{\overset{n\times 1}{\mathbf{v}}} = \mathbf{v} \qquad \overset{n\times n}{\mathbf{I}}\underset{\text{matrix}}{\overset{n\times m}{\mathbf{A}}} = \mathbf{A} \qquad \overset{n\times n}{\mathbf{A}^{-1}}\overset{n\times n}{\mathbf{A}} = \mathbf{I} \qquad \overset{n\times n}{\mathbf{A}}\overset{n\times n}{\mathbf{A}^{-1}} = \mathbf{I}$$

$A \cdot I = A$

by definition of matrix inverse

$A \cdot A^{-1} = I$, $A = A$ ✓

$$\nabla \text{cost}(\mathbf{w}) = -2\mathbf{H}^T(\mathbf{y}-\mathbf{H}\mathbf{w}) + 2\lambda \boxed{\mathbf{w}}$$
$$= -2\mathbf{H}^T(\mathbf{y}-\mathbf{H}\mathbf{w}) + 2\lambda \boxed{\mathbf{I}\mathbf{w}}$$

equivalent

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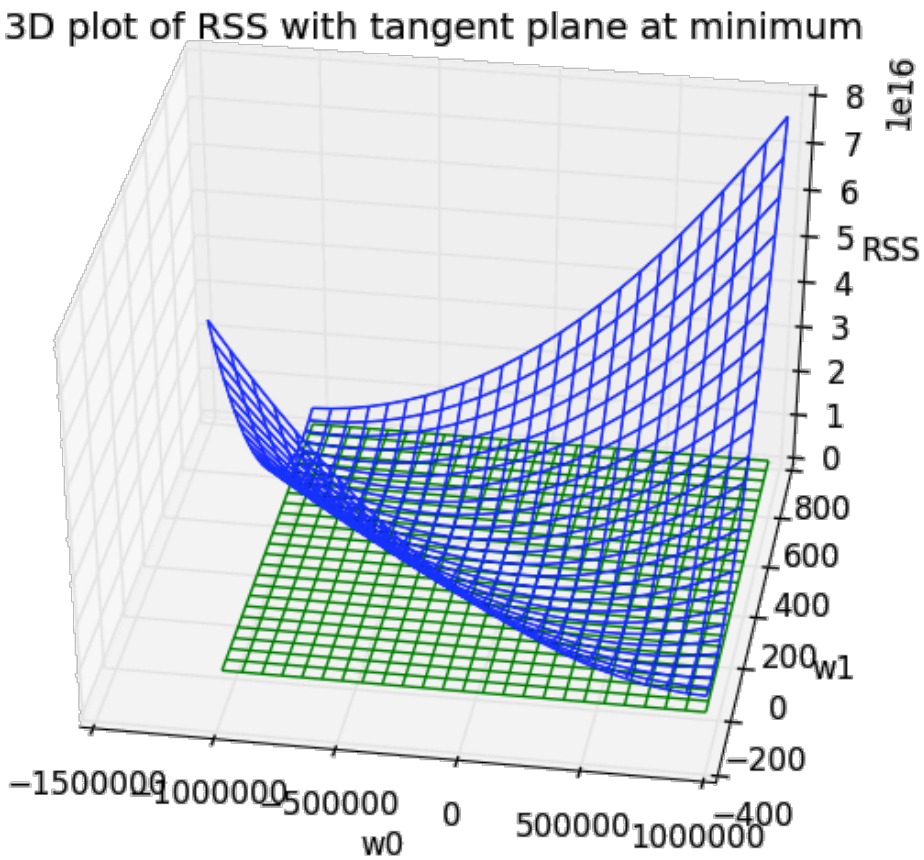
Ridge closed-form solution



$$\nabla \text{cost}(\mathbf{w}) = -\cancel{2}\mathbf{H}^T(\mathbf{y}-\mathbf{H}\mathbf{w}) + \cancel{2}\lambda\mathbf{I}\mathbf{w} = 0$$

Solve for $\mathbf{w}$.

$$-H^T y + H^T H \hat{w} + \lambda I \hat{w} = 0$$

$$H^T H \hat{w} + \lambda I \hat{w} = H^T y$$

$$(H^T H + \lambda I)\hat{w} = H^T y$$

$$\hat{w}^{ridge} = (H^T H + \lambda I)^{-1} H^T y$$

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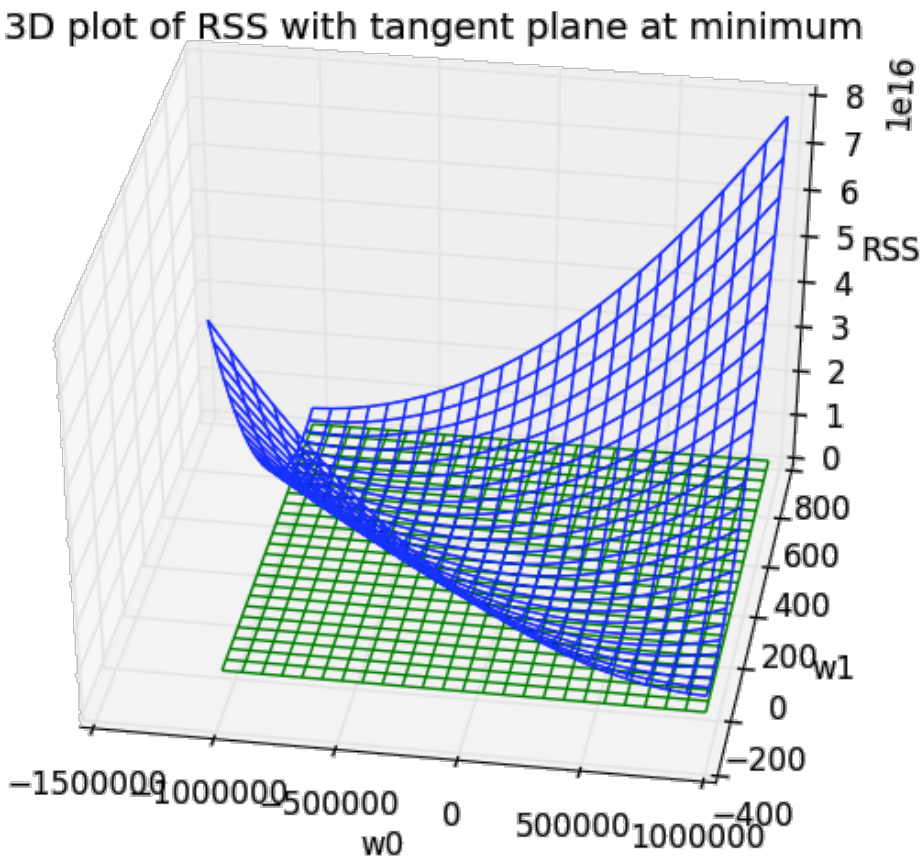
Interpreting ridge closed-form solution



$$\hat{\mathbf{w}}^{ridge} = (\mathbf{H}^T\mathbf{H} + \lambda\mathbf{I})^{-1}\mathbf{H}^T\mathbf{y}$$

If $\lambda=0$: $\hat{w}^{ridge} = (H^TH)^{-1}H^Ty = \hat{w}^{LS}$ ← old solution!

If $\lambda=\infty$: $\hat{w}^{ridge} = 0$ ← because it's like dividing by ∞

✓

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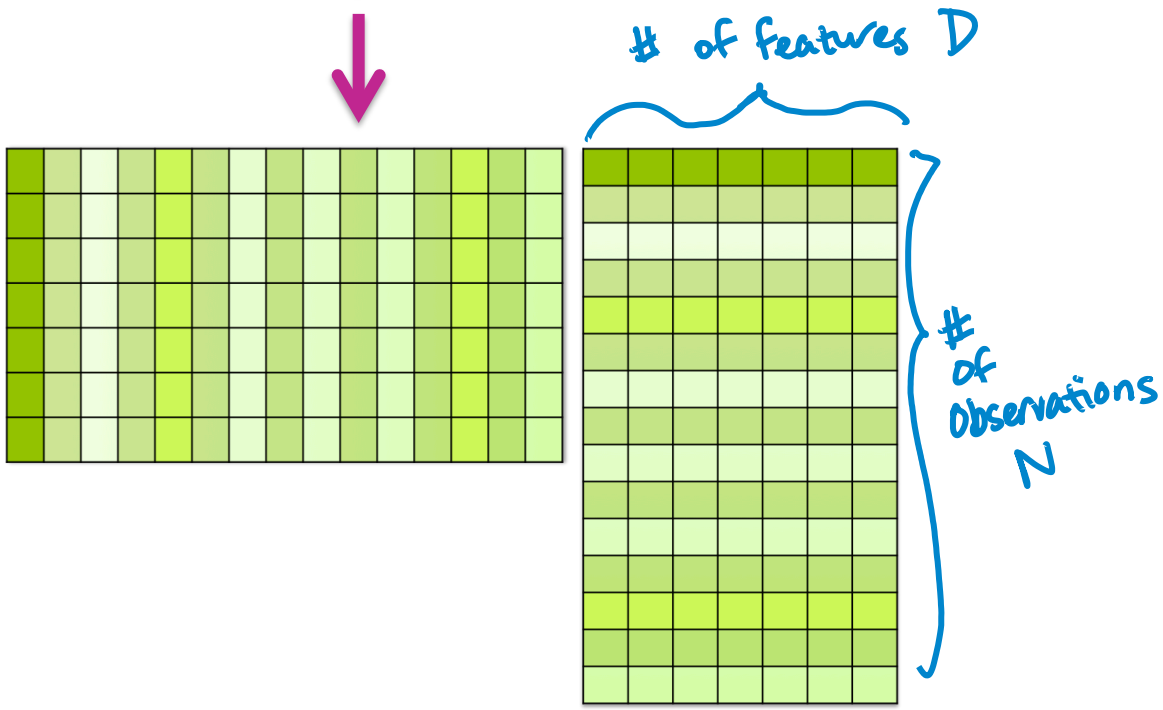
Recall discussion on previous closed-form solution

$$\hat{\mathbf{w}}^{LS} = (\underbrace{\mathbf{H}^T\mathbf{H}}_{})^{-1}\mathbf{H}^T\mathbf{y}$$



Invertible if:

In general,

(# linearly independent obs)

N > D

Complexity of inverse:

$O(D^3)$

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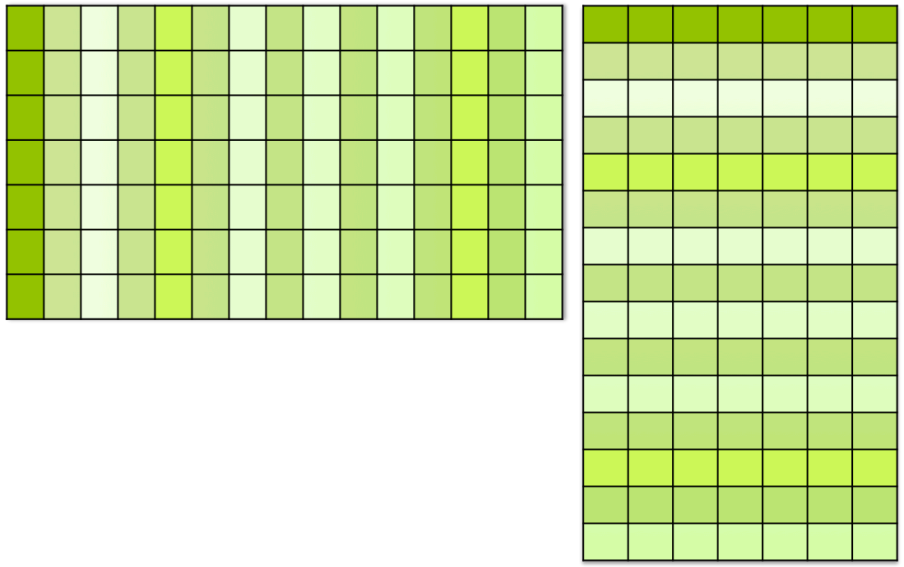
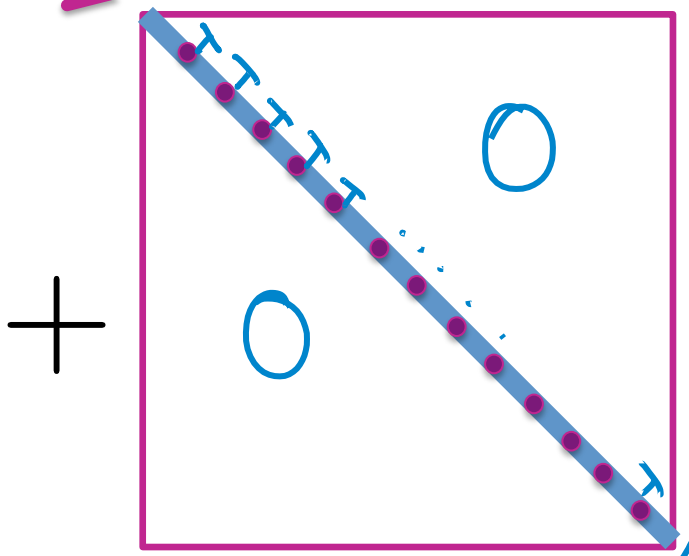
Discussion of ridge closed-form solution

$$\hat{\mathbf{w}} = (\mathbf{H}^T\mathbf{H} + \lambda\mathbf{I})^{-1}\mathbf{H}^T\mathbf{y}$$





really important for large D (lots of features)

Invertible if:

Always if $\lambda>0$, even if $N < D$

Complexity of inverse:

$O(D^3)$...

big for large D!

λI is making $H^TH+\lambda I$ more "regular" → "regularized"

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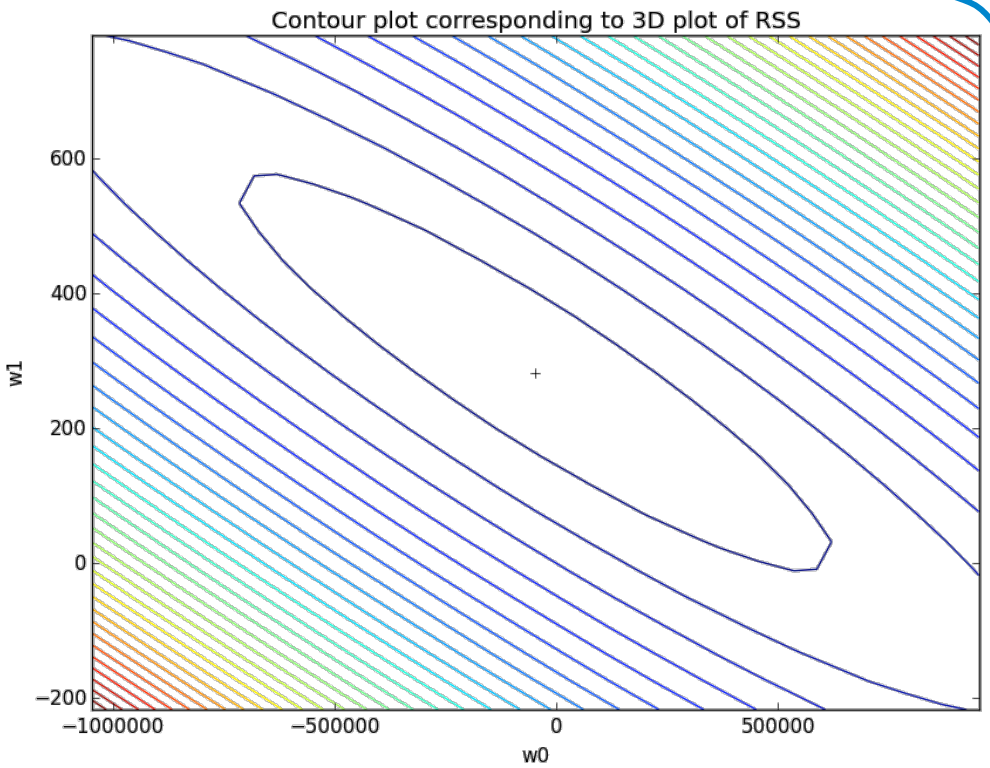
Step 3, Approach 2:
Gradient descent

Elementwise ridge regression gradient descent algorithm

$$\nabla \text{cost}(\mathbf{w}) = -2\mathbf{H}^T(\mathbf{y}-\mathbf{H}\mathbf{w}) + 2\lambda\mathbf{w}$$



Update to j^{th} feature weight:

$$w_j^{(t+1)} \leftarrow w_j^{(t)} - \eta * \left[-2 \sum_{i=1}^{N} h_j(\mathbf{x}_i)(y_i - \hat{y}_i(\mathbf{w}^{(t)})) + 2\lambda w_j^{(t)}\right]$$

Same as before (from RSS term)

new term, comes from the jth component of $2\lambda w$

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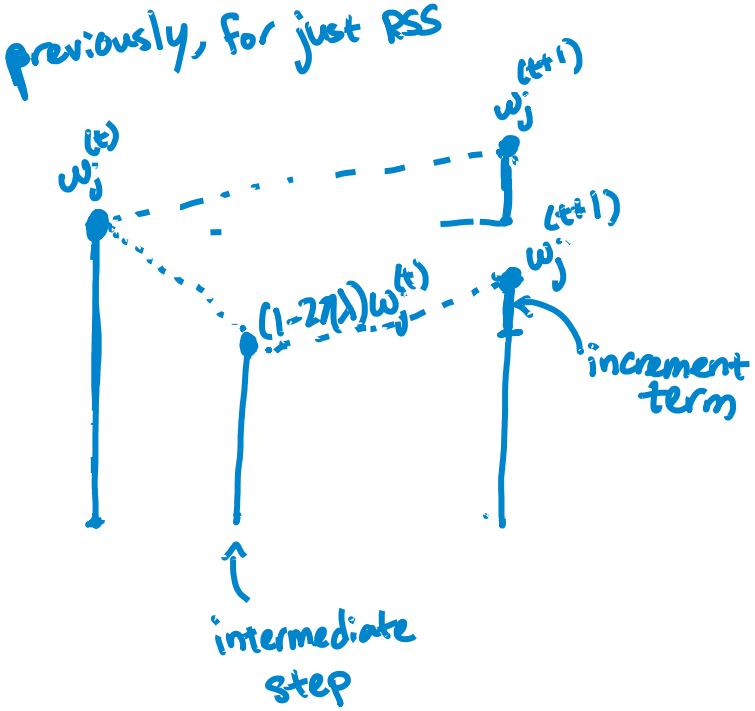
Elementwise ridge regression gradient descent algorithm

$$\nabla \text{cost}(\mathbf{w}) = -2\mathbf{H}^T(\mathbf{y}-\mathbf{H}\mathbf{w}) + 2\lambda\mathbf{w}$$



Equivalently:

$$w_j^{(t+1)} \leftarrow \underbrace{(1-\overbrace{2\eta\lambda}^{2\eta\lambda<1})}_{\leq 1} w_j^{(t)} + 2\eta \underbrace{\sum_{i=1}^{N} h_j(\mathbf{x}_i)(y_i-\hat{y}_i(\mathbf{w}^{(t)}))}_{\text{increment term}}$$

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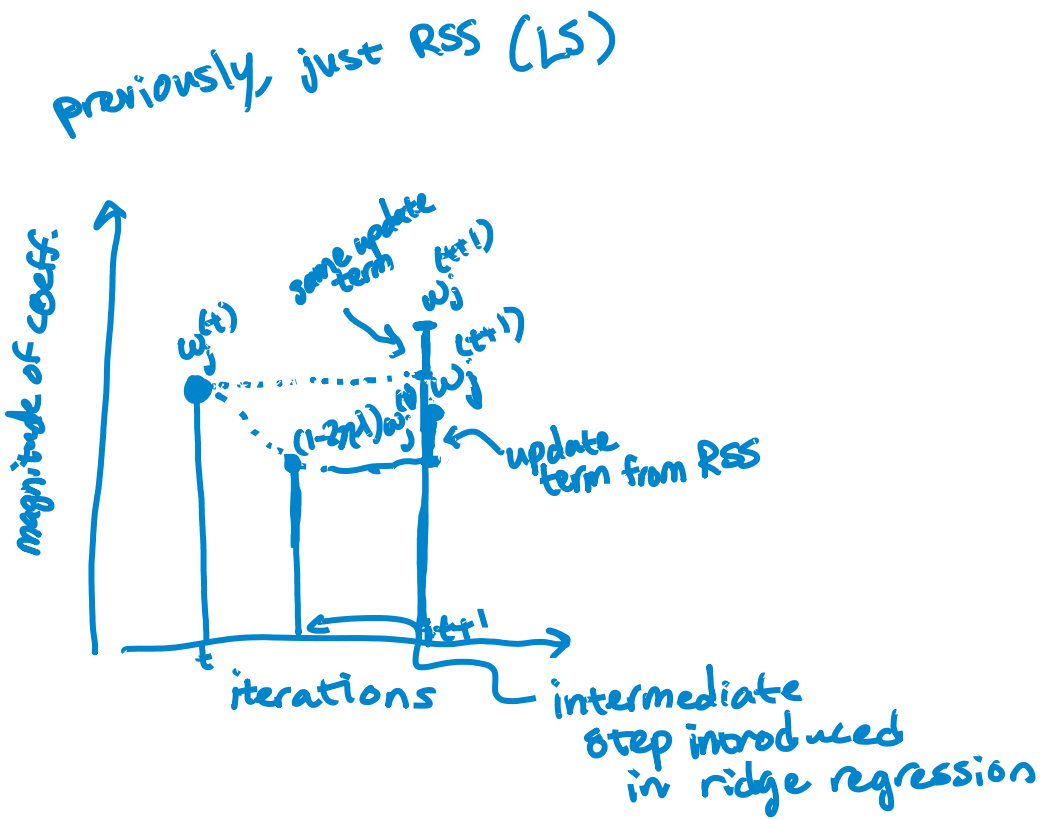
Elementwise ridge regression gradient descent algorithm

$$\nabla \text{cost}(\mathbf{w}) = -2\mathbf{H}^T(\mathbf{y}-\mathbf{H}\mathbf{w}) + 2\lambda\mathbf{w}$$



Equivalently:

$$w_j^{(t+1)} \leftarrow \underbrace{(1-\overbrace{2\eta\lambda}^{2\eta\lambda<1})}_{\leq 1} w_j^{(t)} + \underbrace{2\eta \sum_{i=1}^{N} h_j(\mathbf{x}_i)(y_i - \hat{y}_i(\mathbf{w}^{(t)}))}_{\text{update term from RSS}}$$

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Recall previous algorithm

init $\mathbf{w}^{(1)}=0$ (or randomly, or smartly), $t=1$

while $||\nabla RSS(\mathbf{w}^{(t)})|| > \varepsilon$

for $j=0,\ldots,D$

partial[j] $= -2 \sum_{i=1}^{N} h_j(\mathbf{x}_i)(y_i - \hat{y}_i(\mathbf{w}^{(t)}))$

$w_j^{(t+1)} \leftarrow w_j^{(t)} - \eta$ partial[j]

$t \leftarrow t + 1$

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Summary of ridge regression algorithm

init $\mathbf{w}^{(1)}=0$ (or randomly, or smartly), $t=1$

while $||\nabla RSS(\mathbf{w}^{(t)})|| > \varepsilon$

for $j=0,\ldots,D$

partial[j] $= -2 \sum_{i=1}^{N} h_j(\mathbf{x}_i)(y_i - \hat{y}_i(\mathbf{w}^{(t)}))$

$w_j^{(t+1)} \leftarrow (1-2\eta\lambda)w_j^{(t)} - \eta$ partial[j]

$t \leftarrow t + 1$

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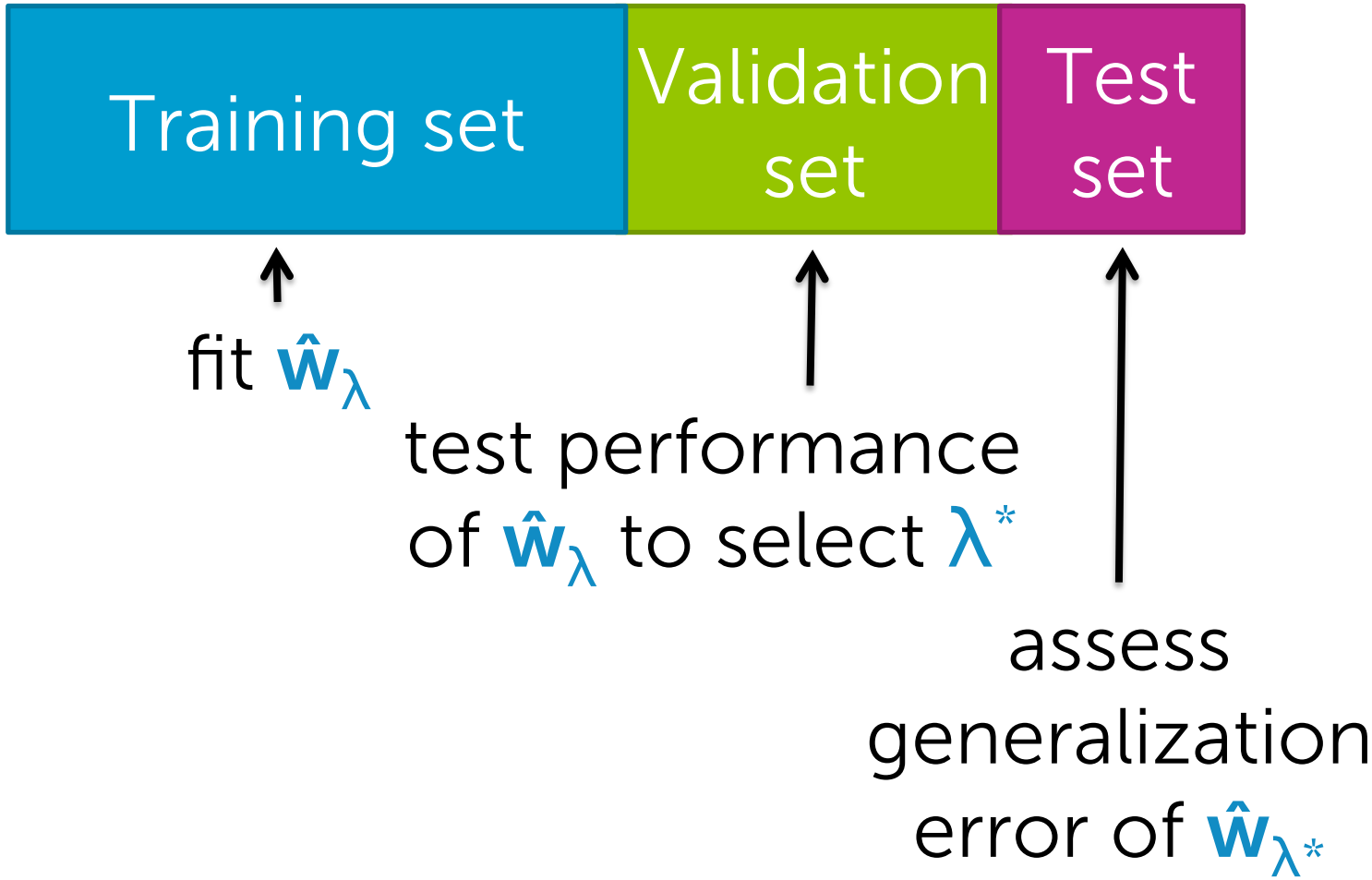
How to choose λ

If sufficient amount of data…



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Start with smallish dataset

All data

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Still form test set and hold out

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How do we use the other data?

Rest of data

use for both training and validation, but not so naively

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Recall naïve approach

small validation set

Is validation set enough to compare performance of $\hat{\mathbf{w}}_\lambda$ across λ values?

No

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Choosing the validation set

small validation set

Didn't have to use the last data points tabulated to form validation set

Can use any data subset

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Choosing the validation set

Which subset should I use?

ALL!

average performance over all choices

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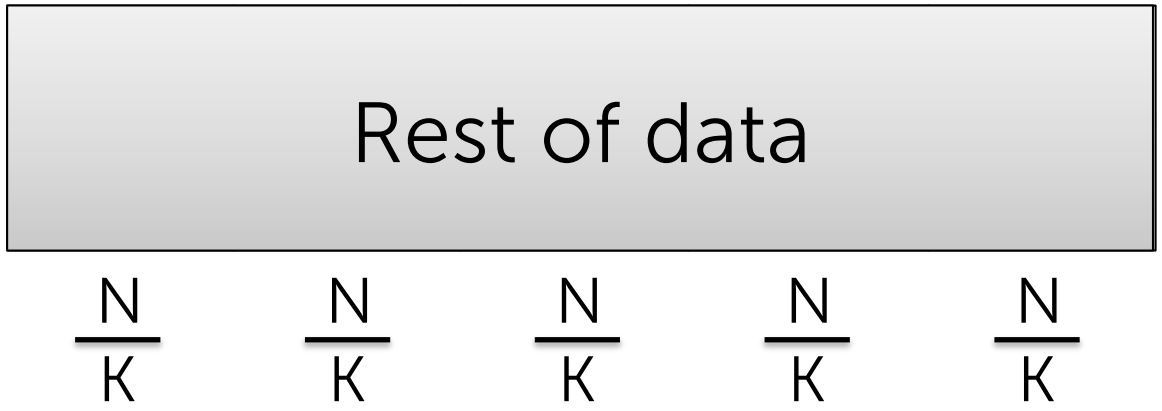
K-fold cross validation



Preprocessing:

Randomly assign data to K groups

(use same split of data for all other steps)

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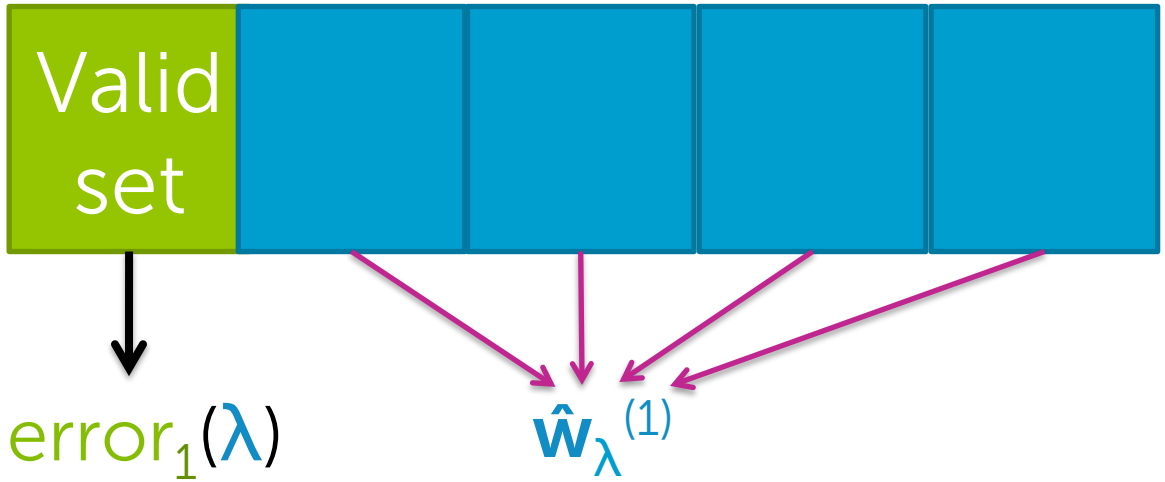
K-fold cross validation



For k=1,…,K

1. Estimate $\hat{\mathbf{w}}_\lambda^{(k)}$ on the training blocks
2. Compute error on validation block: $\text{error}_k(\lambda)$

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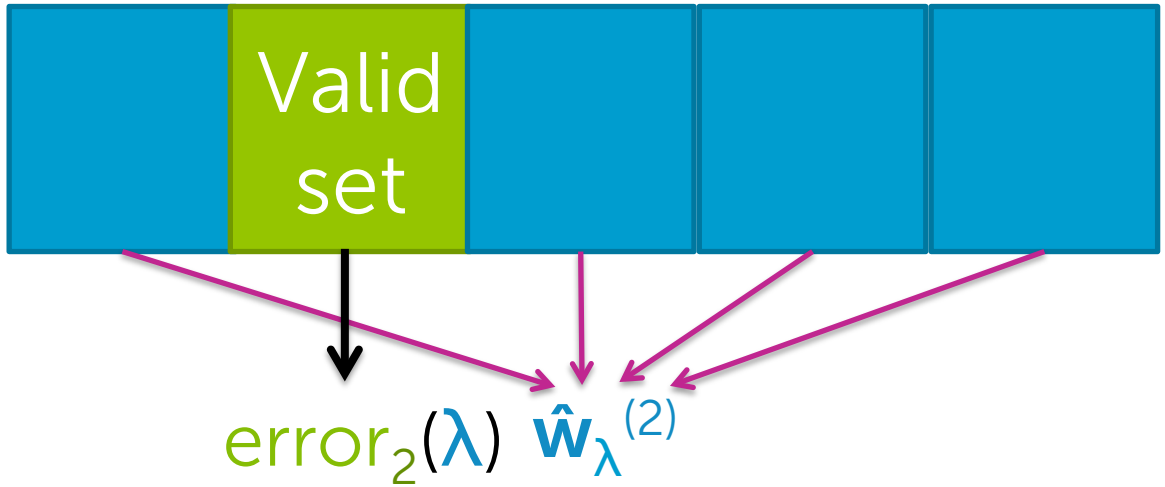
K-fold cross validation



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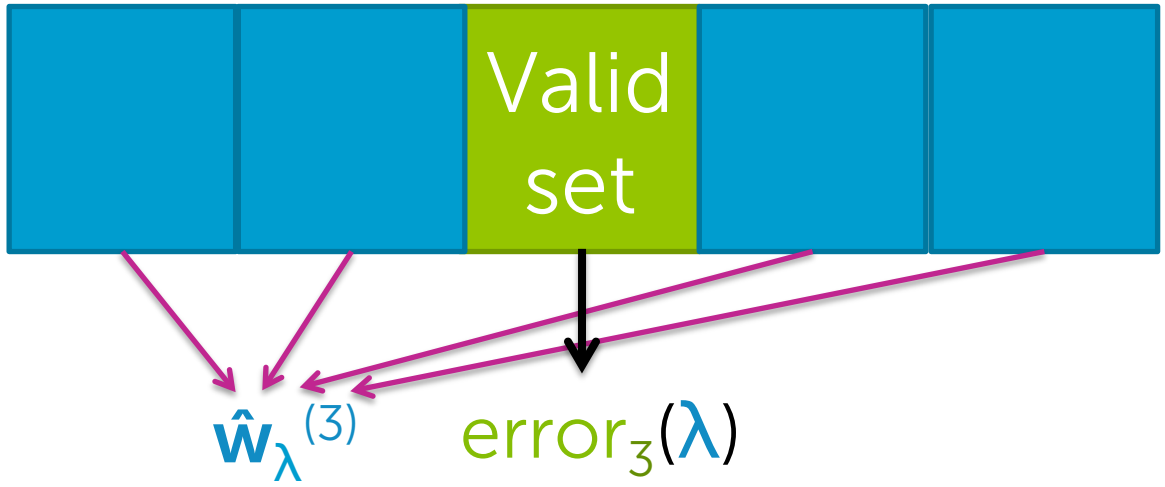
K-fold cross validation



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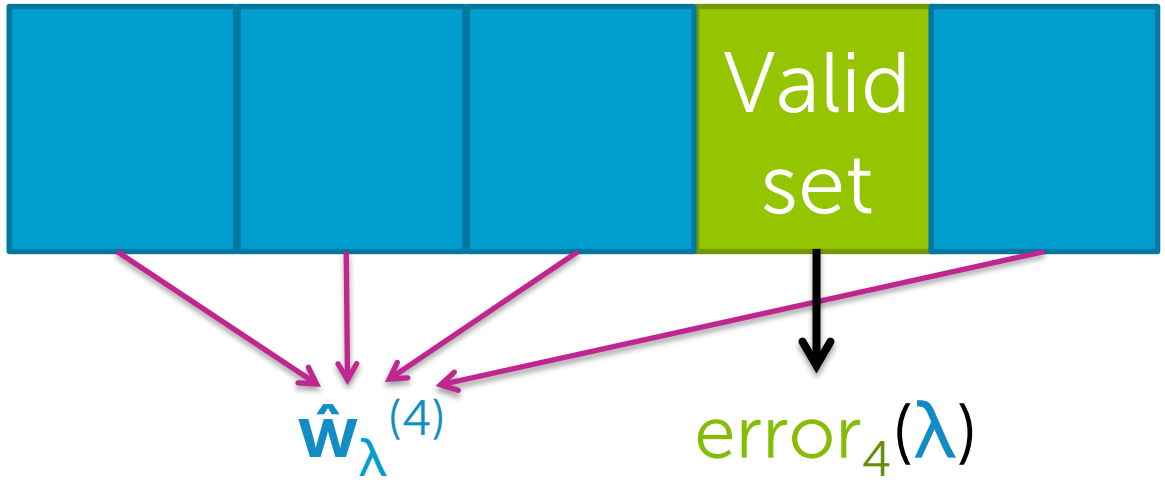
K-fold cross validation



For k=1,…,K

1. Estimate $\hat{\mathbf{w}}_\lambda^{(k)}$ on the training blocks
2. Compute error on validation block: $\text{error}_k(\lambda)$

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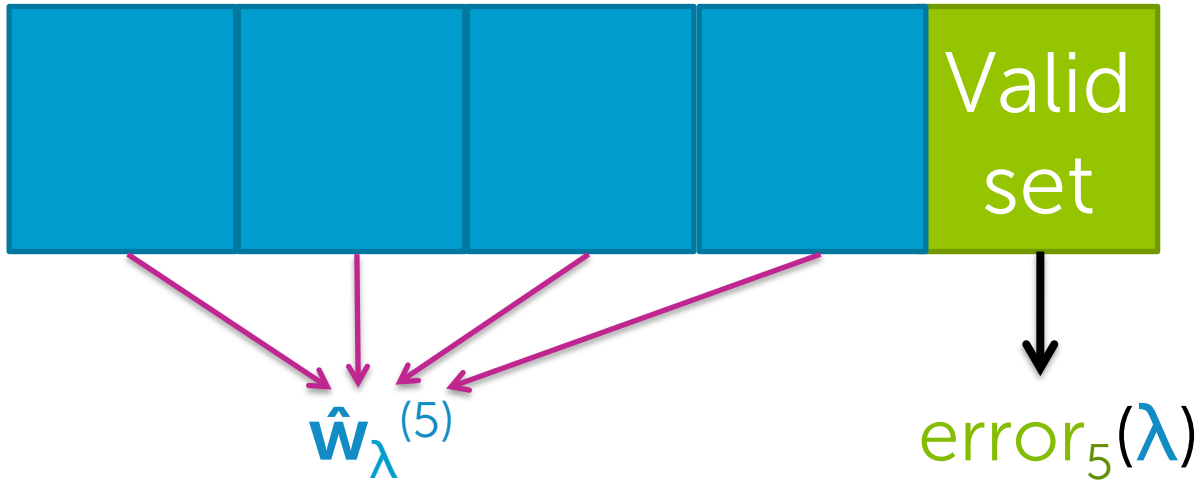
K-fold cross validation



For k=1,…,K

1. Estimate $\hat{\mathbf{w}}_\lambda^{(k)}$ on the training blocks
2. Compute error on validation block: $\text{error}_k(\lambda)$

Compute average error: $\text{CV}(\lambda) = \frac{1}{K}\sum_{k=1}^{K} \text{error}_k(\lambda)$

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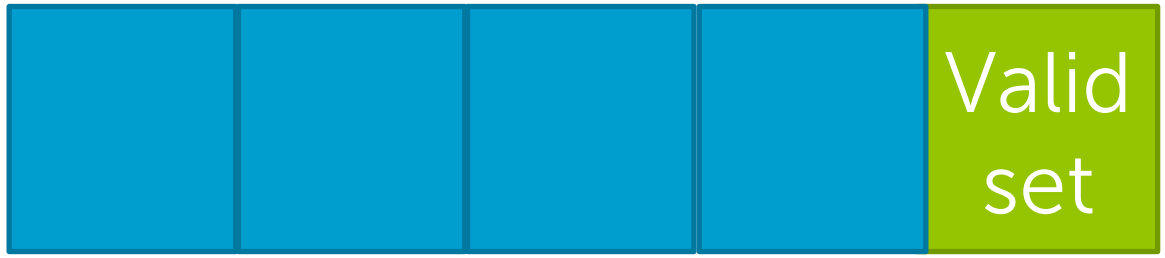
K-fold cross validation



Repeat procedure for each choice of λ

Choose λ^* to minimize $CV(\lambda)$

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What value of K?

Formally, the best approximation occurs for validation sets of size 1 (K=N)

leave-one-out cross validation

Computationally intensive

- requires computing N fits of model per λ

Typically, K=5 or 10

5-fold CV

10-fold CV

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How to handle the intercept

Recall multiple regression model

Model:

$$y_i = w_0 h_0(\mathbf{x}_i) + w_1 h_1(\mathbf{x}_i) + \ldots + w_D h_D(\mathbf{x}_i) + \varepsilon_i$$

$$= \sum_{j=0}^{D} w_j h_j(\mathbf{x}_i) + \varepsilon_i$$

feature 1 = $h_0(\mathbf{x})$…often 1 (constant)

feature 2 = $h_1(\mathbf{x})$… e.g., $\mathbf{x}[1]$

feature 3 = $h_2(\mathbf{x})$… e.g., $\mathbf{x}[2]$

…

feature D+1 = $h_D(\mathbf{x})$… e.g., $\mathbf{x}[d]$

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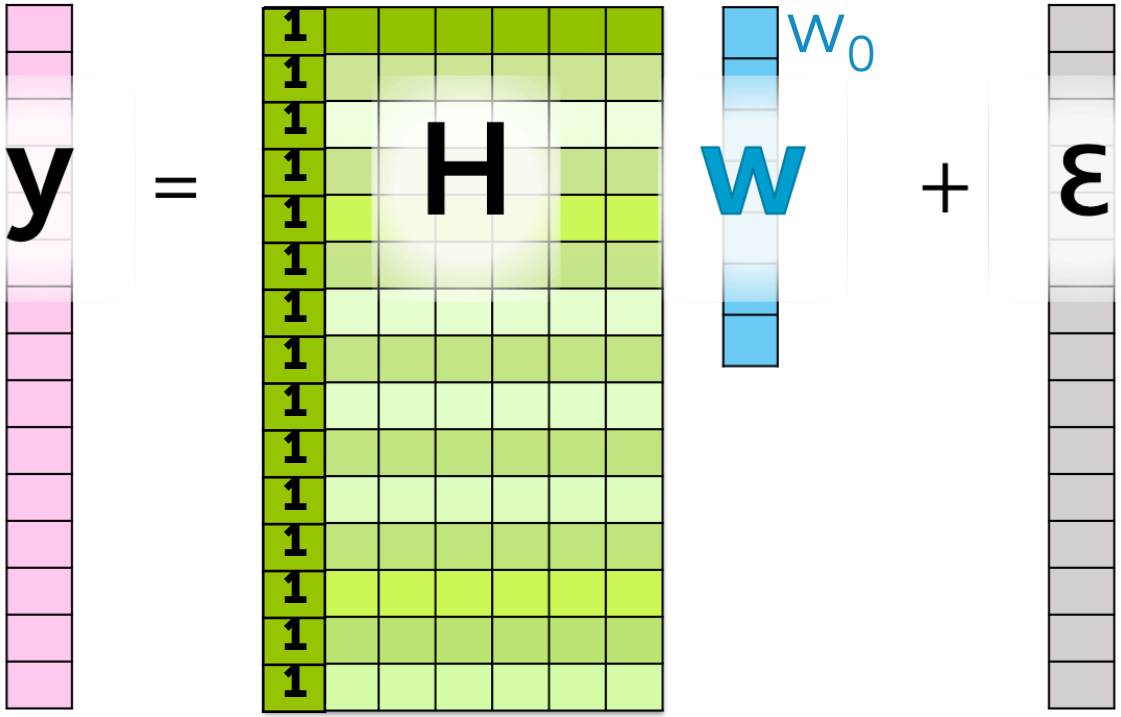
If constant feature…

$$y_i = w_0 + w_1 h_1(\mathbf{x}_i) + \ldots + w_D h_D(\mathbf{x}_i) + \varepsilon_i$$

In matrix notation for N observations:



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Do we penalize intercept?

Standard ridge regression cost:

$$\text{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|_2^2$$

strength of penalty

Encourages intercept w_0 to also be small

Do we want a small intercept?
Conceptually, not indicative of overfitting…

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Option 1: Don't penalize intercept

Modified ridge regression cost:

$$\text{RSS}(w_0, \mathbf{w}_{rest}) + \lambda \|\mathbf{w}_{rest}\|_2^2$$

How to implement this in practice?

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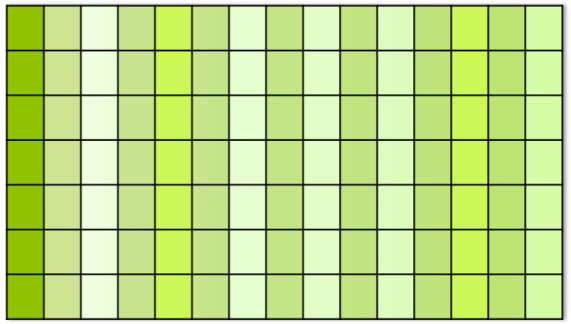
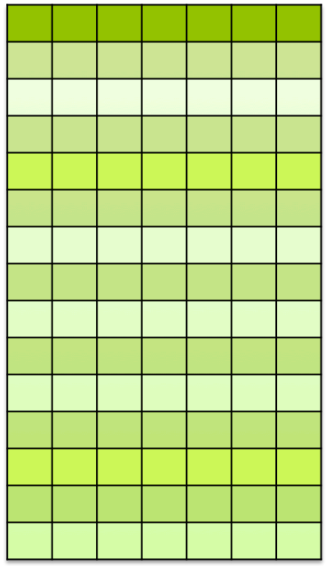
Option 1: Don't penalize intercept – Closed-form solution –

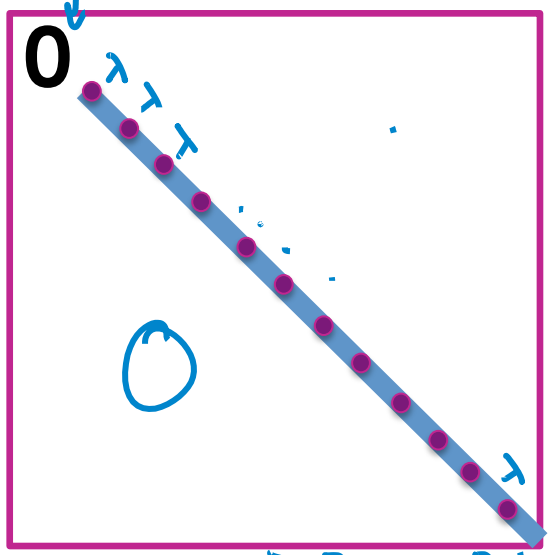
$$\hat{\mathbf{w}} = (\underbrace{\mathbf{H}^T\mathbf{H}} + \underbrace{\lambda\mathbf{I}^{mod}})^{-1}\mathbf{H}^T\mathbf{y}$$





+

no penalty in entry corresponding to w_0 index



new penalty:

$\lambda\, w_{rest}^T w_{rest}$

gradient:

$2\lambda\, w_{rest} = 2\lambda \overset{(D-1)\times(D-1)}{I}\, \overset{D-1\times 1}{w_{rest}} \rightarrow 2\lambda \overset{D\times D}{\begin{bmatrix} 0 & & & \\ & 1 & & \\ & & 1 & \\ & & & \ddots \\ & & & & 1 \end{bmatrix}} \overset{D\times 1}{\begin{bmatrix} w_0 \\ w_{rest} \end{bmatrix}}$

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Option 1: Don't penalize intercept – Gradient descent algorithm –

while $||\nabla RSS(\mathbf{w}^{(t)})|| > \varepsilon$

for j=0,…,D

partial[j] $= -2 \sum_{i=1}^{N} h_j(\mathbf{x}_i)(y_i - \hat{y}_i(\mathbf{w}^{(t)}))$

if j==0

$w_0^{(t+1)} \leftarrow w_0^{(t)} - \eta$ partial[j] ← old LS update (no shrinkage to w_0)

else ← for all other features

$w_j^{(t+1)} \leftarrow (1-2\eta\lambda)w_j^{(t)} - \eta$ partial[j] ← ridge update

$t \leftarrow t + 1$

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Option 2: Center data first

If data are first centered about 0, then favoring small intercept not so worrisome

Step 1: Transform y to have 0 mean

Step 2: Run ridge regression as normal

(closed-form or gradient algorithms)

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Summary for ridge regression

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What you can do now…

- Describe what happens to magnitude of estimated coefficients when model is overfit
- Motivate form of ridge regression cost function
- Describe what happens to estimated coefficients of ridge regression as tuning parameter λ is varied
- Interpret coefficient path plot
- Estimate ridge regression parameters:
  - In closed form
  - Using an iterative gradient descent algorithm
- Implement K-fold cross validation to select the ridge regression tuning parameter λ

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